Controlling Chaos

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Why control of chaos is needed

Chaos is a dynamical system defined as systems which have a chaotic orbit defined as -

. aperiodic, bounded, positive “Lyapunov exponents = lim Sum(ln(|f’(xi)|)/n”. The w limit set of such systems are typically fractal in structure, and they often contain infinite number of unstable periodic orbits, densely. * Devanay’s definition is transitivity, infinitely dense set of UPO’s, sensitivity

-> Stabilizing by turning into a non chaotic system may be too difficult

-> As chaos is highly sensitive, control may be done with only small signals

-> Chaotic systems typically have an infinite number of unstable periodic orbits, and we may want to change behaviour of a system just by switching orbit. This is hypothesised to be a factor in how the brain works, and leads to neuromorphic computing studies.

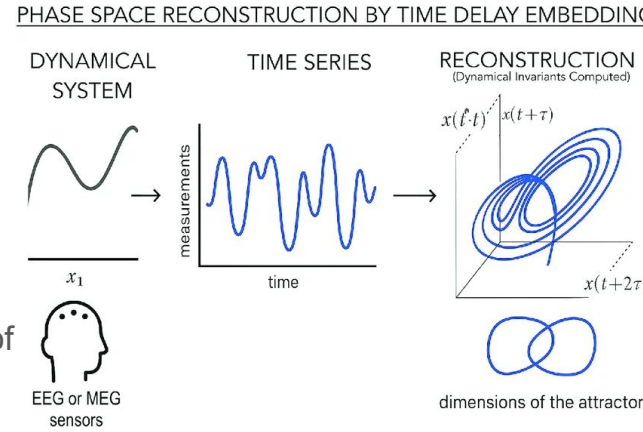
Key ideas

Challenges - System is unknown, imprecise/noisy control and system, only 1 parameter “p” can be changed, so dx/dt = F(x,p) is the system model.

Step 1 - extract an image of the chaotic attractor using “time series embedding”.

Do this for multiple parameters p

Taken’s embedding theorem states that for a manifold M, and a f:M->M be a map, where d is the dimension of the manifold, and a h:M->R, the vector whose components are h(p),h(f(p))...h(f^(2*d)(p)) over all p is an embedding of M in R^(2d+1). This means if f can be taken as the flow of a system after time T, the “time series embedding” caused by measuring a physically accessible observable h in steps of T starting at t, is diffeomorphic to the real phase space of the system, and specifically when the system is moving along/near to its attractor, we get a picture of it’s attractor.

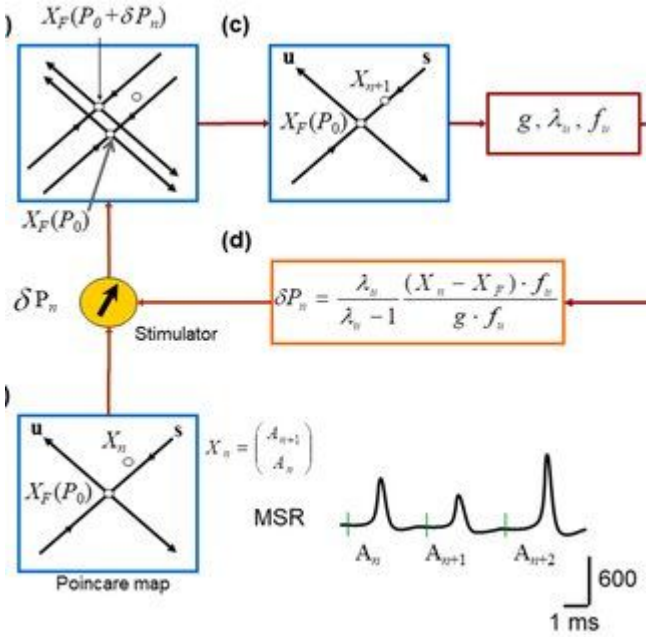


Step 2 -> Identify a periodic orbit of the require degree, there are a few ways to do this, one is a brute force computation to see which curves appear closed for a long time. The 2nd is to construct a poincare section at some point, and use a root finding algorithm like Newton-Raphson to find x s.t. f^n(x)-x is 0.

Step 3 -> Using Anasov closing lemma, there will always exist a real periodic orbit arbitrarily close to this experimentally found near periodic looking orbit. (every delta-pseudo orbit is epsilon shadowed by a true orbit (terms defined and proved later)). Now pick the orbit you want stabilized, at p=0 (no control p). Construct it's poincare return map at a certain transversal (orbit is perpendicular to plane) plane slice through a point on the orbit, x.

x_{n+1} - x = A*(x_n - x) + B*dp, for that poincare map, with A,B experimentally reconstructed.

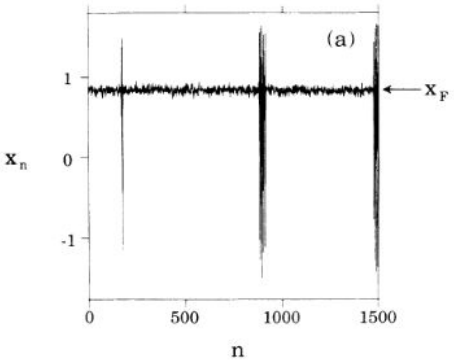
Step 4 -> Now keep on observing the system, whenever it comes close to (on, if we can measure precisely enough) the poincare map, adjust p such that the deviation of the system from the periodic point on the poincare return map is along the stable manifold of F(x,dp). Now by the time the point returns, it would have come closer to the origin due to being on stable manifold and p can return to being 0, if noise kicks it away, repeat by adjusting p. This will ensure that unless the noise becomes too large, the system stays on the periodic orbit.



Control law-

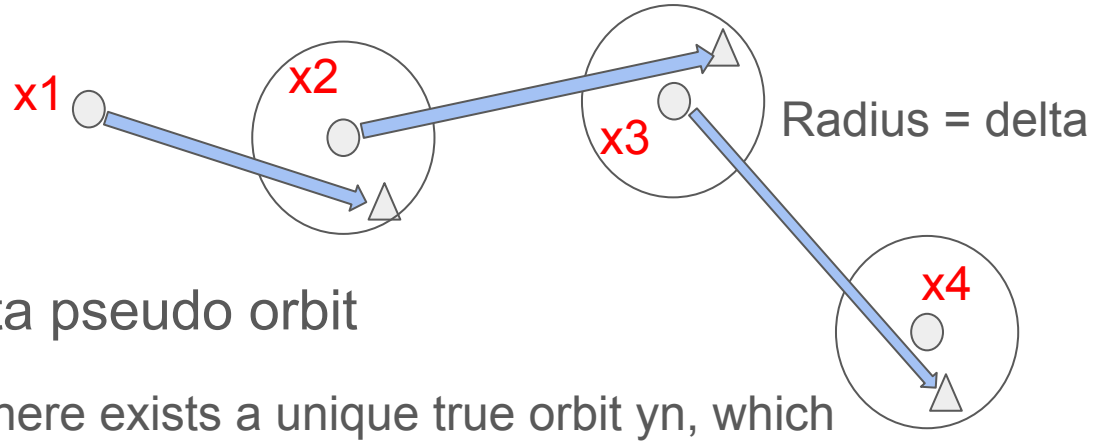
0 = fu.[A(x_n - x_F) + B*dp] , where fu is unstable eigenvector of A, and B represents drift in orbit due to dp.
dp = -lambda_u *fu.(x_n - x_F)/(fu.B) , where lambda_u is unstable eigenvalue (its modulus greater than 1)

1) This is a “stroboscopic” not a continuous control method
2) This serves as a proof of concept for chaos control more than anything else, more modern techniques have been developed in the 30 years since, some techniques include Pyragas delayed feedback method [apply feedback proportional at each time t
3) to x(t)-x(t-T) , where T is one time period.], ML/DL based controllers
4) If a noise kicks the system too far from the orbit, instead of
5) controlling it, just wait until the system comes close to the orbit again and apply control once more, this is guaranteed to happen in a finite time (given by kac’s theorem)
6) All this theory applies without much hassle to “uniformly hyperbolic” systems, rigorous results for general chaotic systems is an open problem.
7) The recurrence time, after which a system that has lost control returns close to the periodic orbit may be impractically large, nor can it in general be bounded deterministically by maths.
8) In addition, topological transitivity is required for recovery after a noise pulse



Example system simulated in the paper, showing effect of noise

Shadowing Lemma



Delta pseudo orbit

There exists a unique true orbit yn, which is always within some epsilon of xn, if the dynamics is uniformly hyperbolic (uniform hyperbolic means hyperbolic at all points, with bounds on contraction rate and angle between stable and a=unstable manifolds)

Definition of hyperbolicity -
With steps, unstable manifolds map to unstable manifolds and vice versa for stable manifolds
Also, Jn = Df^n always splits into 2 subspaces, with one subspace contracting by v^(-n) in forward direction, other by some u^(-n) in backward direction.

Topological Expansiveness

There is a constant alpha such that, for any 2 points that are arbitrarily close, after some N number of iterations, either forwards or backwards, the distance between the points will cross alpha. All hyperbolic maps are topologically expansive.

Anasov Closing lemma

If f^n(x) returns to within beta of x, then for some alpha, there is a n-period orbit within Ball(c,alpha).

Existence of infinite number of periodic orbits follows from this, consider any ball at any point, by topological transitivity, there exists n such that f^n(point) returns to the ball, now use Anasov’s closing lemma to construct a periodic orbit.

Whitney Embedding Theorom

An immersion is a map f: M -> N such that the rank of the matrix of partial derivatives of the map is of constant rank. An embedding is an injective immersion that is also a diffeomorphism.

Whitney's theorem states that there always exists a diffeomorphism from M to R^(2d+1) where d is the dimension of the manifold M. The intuition behind this theorem is that if we take 2 separate open subsets of M, both homeomorphic to a domain of R^d, they will intersect at atleast on a 2d-N dimensional surface. For N=2d, it is like 2 dimensional planes that intersect at a point, and for N=2d+1, intersection can be avoided.

Taken's Theorom

Taken's theorem gives a construction of a Whitney type embedding.f:M->M be a map, where d is the dimension of the manifold, and a h:M->R, the vector whose components are h(p),h(f(p))...h(f^(2*d)(p)) over all p is an embedding of M in R^(2d+1).

Understanding henon map (a non hyperbolic example)

$x_{n+1} = 1-A*xn^2+B*yn$

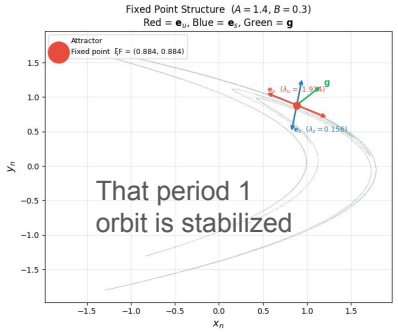
$Y_{n+1} = xn$

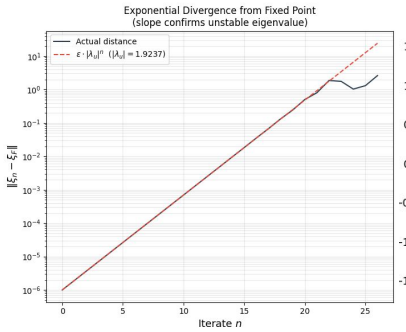
$DH(x,y)=((-2Ax,1),(B,0))$ is the Jacobian, whose determinant is -B

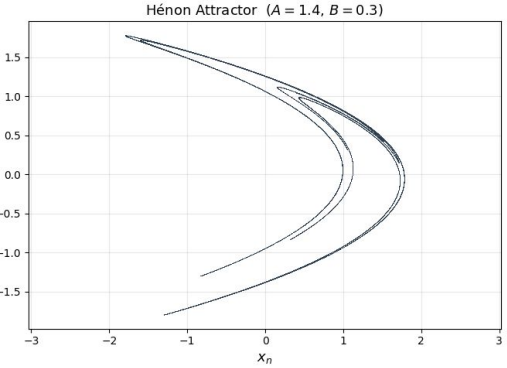
Michael Benedicks and Lennart Carleson (1991)

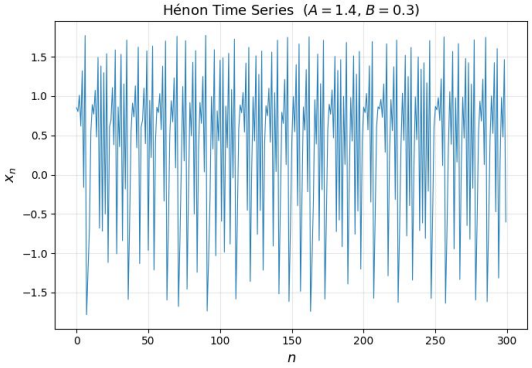
Proved that near (A,B)=(2,0) there is a finite measure set that is nonuniformly hyperbolic, has a strange attractor, behaves chaotically. The pair won the Abel prize for that work.

Simulated Henon map control

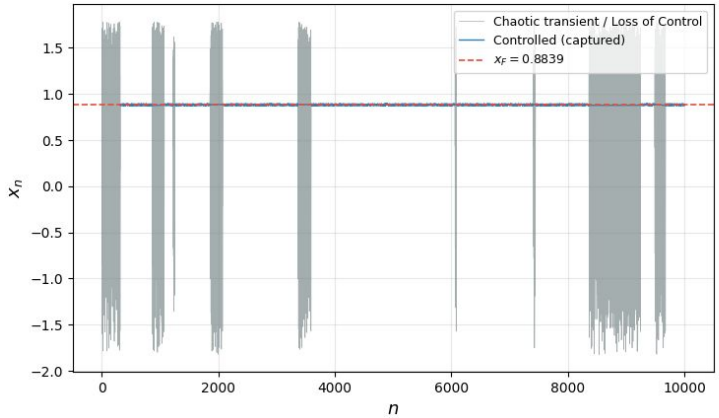


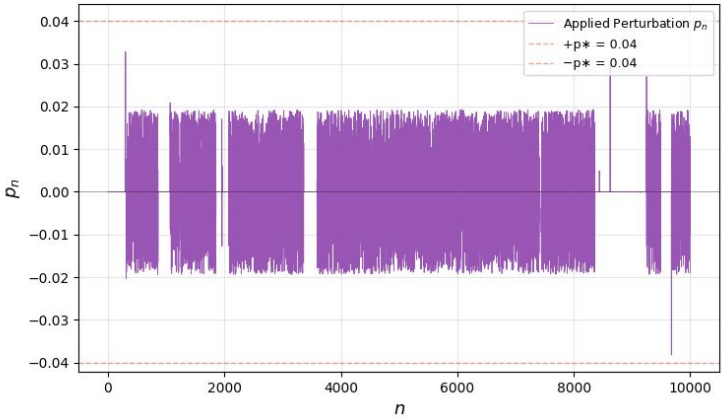






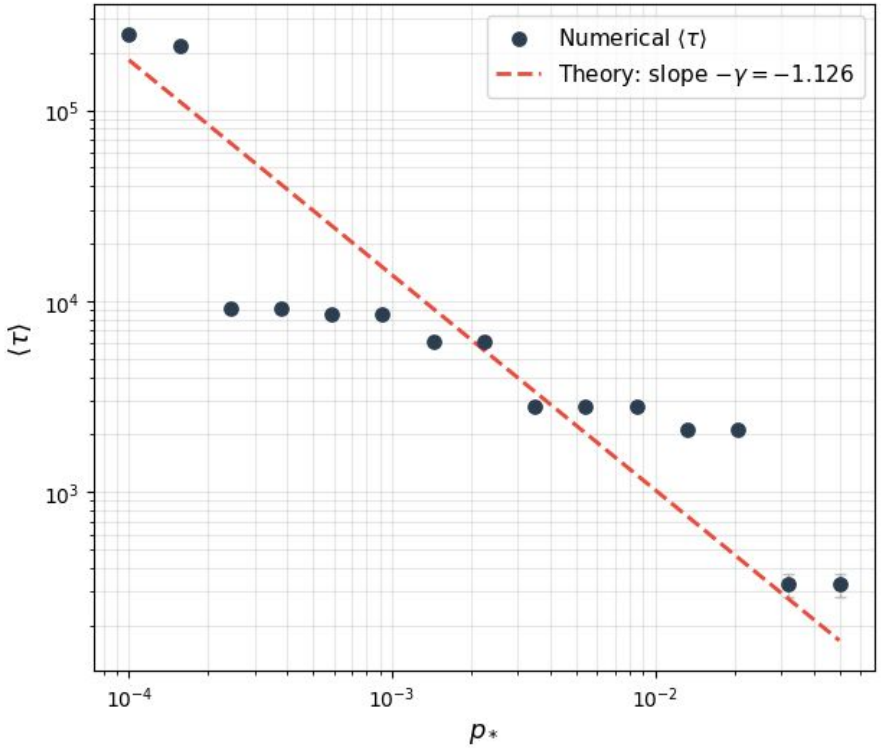
OGY Control Under Rare Poisson Shocks (p_max=0.04, Shock Jump=0.5, Probability=1/1000)





Numerical simulation

The poincare section is assumed to be a henon map (many 3 state variable physical systems have henon like behaviour) in this paper. The average time for recapture vs the max displacement before losing control is plotted in log-log form. A=1.4, B=0.3 is used.0.2 is the distance displaced on poincare map by a noise, and p* is the capture bounds. <T> vs noise follows a power law.



Conclusion

- We have seen the first technique of control of chaos which was developed.
- After using time series embedding to reconstruct a picture of the chaotic attractor, a poincare section is made at a point near a selected periodic orbit.
- Whenever the system comes on that attractor, it's displacement from the chosen periodic point is found, and a parameter that is accessible, is changed in a step function manner so as to make that displacement be on the stable manifold of the poincare map. This will keep the system close to the unstable periodic orbit, as long as it doesn't wander too far away.
- Finally this technique was tested in a simulation for a Henon map with A=1.4, B=0.3 to stabilize a period 1 orbit.